

LIANG YUZHE

Telephone number: +86 19986662829
Email Address: lyz123456@mail.ustc.edu.cn

EDUCATION

University of Science and Technology of China 2023 – 2027 (Expected)

- School of Management
- Major: Statistics
- GPA: 3.56/4.3(Rank:7/36)

Course work: Mathematical Analysis I (92), Mathematical Analysis II (95), College Physics II (99), Probability (90), Functional Analysis (92), Regression Analysis (92)

SCHOLARSHIPS AND AWARDS

Freshman Scholarship, USTC	2023
Outstanding Student Scholarship (Bronze Award), USTC	2024
Outstanding Student Scholarship (Bronze Award), USTC	2025
National College Mathematics Competition (Provincial Third Prize)	2023
Outstanding Student Representative, Future Technology Leaders Research Camp, Nagoya, Japan	2025

PROJECTS

Flow Matching for Data Assimilation in Atmospheric Sciences (Research Project, advised by Prof. Catherine Chunling Liu, The Hong Kong Polytechnic University; Jan. 2026 – Present)

- Project Description: Developing a Flow Matching-based amortized posterior sampler for data assimilation, where MCMC samplers (HMC / MALA) generate (prior, posterior) sample pairs offline to fine-tune a pre-trained generative backbone into a one-step analysis-field generator.
- My work (conducted independently under the guidance of the advisor and a postdoctoral mentor):
 - Independently derived the theoretical framework, including the Bayesian formulation and the Flow Matching fine-tuning objective.
 - Implemented HMC / MALA samplers and completed systematic MALA tuning on a simulated benchmark.
 - Three-month on-site research at PolyU in Summer 2026; manuscript targeted for journal submission by end of 2026.

Financial Time Series Volatility Breakpoint Detection under a Bayesian Framework (Course Project)

- Project Description: A Bayesian Metropolis-Hastings framework for detecting volatility change points in financial time series under external shocks.
- My work:

- Cleaned and preprocessed the 2008 S&P 500 dataset; implemented the R-based MH sampler (likelihood, proposal, acceptance, multi-chain) and tuned step size across priors.
- Produced posterior, trace, and change-point figures with ESS diagnostics; interpreted results, confirming the model identified the regime shift one trading day before the Lehman Brothers bankruptcy.

SKILLS

Programming and Data Analysis

- Proficient in **C**, **Python**, and **R** for data processing, statistical analysis, and modeling.
- Skilled in document preparation using **Markdown** and **LaTeX**.
- Solid foundation in probability theory and mathematical statistics; experienced in solving fundamental mathematical problems.

Languages

TOEFL: 90, CET-6:554

ACTIVITIES

Future Technology Leaders Research Camp, Nagoya, Japan

Winter 2025

Teaching Assistant, Probability Theory and Mathematical Statistics, USTC

Spring 2026